

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a60579fd70e7e8f60.jsonl`  
- SHA-256: `0b58fbab79526593afed8487cb6412362b9ae1f572b0af7442cde677a9619d54`  
- Source modified: `2026-06-16T03:19:49+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:04.329Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/html/2407.17792>

****Title:**** Harnessing Temporal Causality for Advanced Temporal Action Detection (CausalTAD, arXiv 2024)

****Found via:**** Single-stage TAL actionness + boundary heads, small data search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:07.137Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:19:26.171Z)

Technical Summary: CausalTAD for Temporal Action Detection

(1) Core Method & Temporal Causality Modeling

****Method Name:**** CausalTAD combines causal attention and causal Mamba in a hybrid block.

The core innovation addresses a fundamental assumption in prior work. As stated: `"these modules typically treat past and future information equally, overlooking the crucial fact that changes in action boundaries are essentially causal events."`

`**Causal Attention Implementation:**`

The method restricts attention via a causal mask. For backward attention (past-only context):
`"M_{ij} = 0 if i > j, else -1"` This ensures `"each position in the sequence can only attend to its past and present positions, effectively modeling temporal causality."`

`**Causal Mamba Block:**`

Follows decomposed bidirectional design from VideoMambaSuite, with `"input projectors learned separately in each direction, while the parameters in the forward and backward MHSA...are shared."`

(2) Detection Head & Boundary Handling

The paper builds on `**ActionFormer**`, described as `"a simple yet effective one-stage anchor-free method."` It employs:

- Regression loss for boundary prediction (hyperparameter: regression loss weight)
- Channel dropout (hyperparameter: probability of input channel dropout)
- Post-processing with Soft-NMS for redundant proposal elimination

No explicit over-segmentation handling discussed beyond standard NMS with configurable sigma threshold.

(3) Benchmark Results (mAP@tIoU)

`**THUMOS14 (I3D features):**`

- CausalTAD: 84.4@0.3, 80.7@0.4, 73.5@0.5, 62.7@0.6, 47.4@0.7
- `*Average mAP: 69.7%* (vs. TriDet 69.3%)`

`**ActivityNet-1.3 (TSP features):**`

- CausalTAD: 55.62@0.5, 38.51@0.75, 9.40@0.95
- `*Average mAP: 37.46%* (vs. TriDet 36.80%)`

`**EPIC-Kitchens 100 (SlowFast-R50, validation):**`

- Verb task avg: 26.4% mAP
- Noun task avg: 24.3% mAP

`**Ego4D Moment Queries (validation, InternVideo1):**`

- `*Average mAP: 32.19%*`

`**Test Set Results (with feature ensembling):**`

- Ego4D-MQ test: `*34.99% average mAP*`
- EPIC-Kitchens action task test: `*31.97% average mAP*`

No FineAction or HACS results reported.

(4) Single-Stage Architecture

Yes, confirmed: `"Our detection model is based on ActionFormer, a simple yet effective one-stage anchor-free method."`

(5) Architecture Details

`**Hybrid Causal Block Components:**`

- Multi-Head Self-Attention (MHSA) with causal masking
- Mamba/SSM (Structured State-Space Model)
- `"We compute the causal Mamba block and causal attention block separately, concatenate their outputs, and use a linear layer to reduce the channel dimension."`

`**Feature Extraction:**`

Multi-scale feature pyramids within ActionFormer framework; `"sliding window approach to extract`

features for each clip independently."*

(6) Publication Details

****ArXiv ID:**** 2407.17792v2

****Posted:**** July 2024

****Venues:**** EPIC-Kitchens Challenge 2024 (1st place), Ego4D Challenge 2024 (1st place)

****Code:**** Available at <https://github.com/sming256/OpenTAD>

4. assistant (2026-06-16T03:19:32.549Z)

I have the source content. Let me assess and structure the findings.

The source is a primary research paper (arXiv 2407.17792, CausalTAD). It's relevant to Dimension 1 (Windowing & Boundaries / TAL) of the research question. Key relevance: it builds on ActionFormer (single-stage anchor-free actionness + boundary regression), introduces causal modeling of action boundaries, and reports state-of-the-art numbers. However, it's a large-data benchmark method (THUMOS14, ActivityNet, Ego4D, EPIC-Kitchens) with no small-data/tiny-label evaluation, and the over-segmentation handling is only standard Soft-NMS, not the segmental-F1/edit-score-oriented approaches the question asks about.

5. user (2026-06-16T03:19:40.720Z)

Structured output provided successfully

6. assistant (2026-06-16T03:19:49.117Z)

Done. Extracted 5 falsifiable claims from CausalTAD (arXiv 2407.17792).

Source assessment: primary research (arXiv paper, code at github.com/sming256/OpenTAD, challenge-winning method). Relevant to Dimension 1 (Windowing & Boundaries / TAL).

Key findings for the research question:

- Central relevance: It demonstrates a recent (2024) single-stage anchor-free TAL design (ActionFormer-based) and introduces causal boundary modeling ? treating action-boundary changes as causal events via a backward-causal attention mask. This is conceptually on-point for the over-merge problem (boundary-aware regression).
- Important caveat for the constrained setting: The over-segmentation handling is only standard Soft-NMS, not the segmental-F1 / edit-score / explicit boundary-split machinery the question targets (ASRF, MS-TCN, BMN). The architecture uses Mamba + transformer over multi-scale I3D/SlowFast/InternVideo feature pyramids ? heavy, large-data (THUMOS14/ActivityNet/Ego4D/EPIC) with no tiny-label evaluation and relying on feature ensembling. So this is a research-bet/reference baseline, not a low-effort win for a ~6?16 video, 1-D-trajectory setting.

All claims are quote-backed; nothing license-incompatible was asserted (the paper is method-only; I did not verify a license for the OpenTAD code).